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RESEARCH

Radical Perspectivalism, Locality and Relativity

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Abstract
Perspectivalism is a natural ingredient of unitary one-world quantum mechanics. After briefly reviewing arguments for this thesis, we argue that a radical version of perspectivalism is able to provide local and relativistically covariant accounts of physical processes, and thus offers a way out of several no-go theorems. According to this radical perspectivalism, different perspectives are independent of each other and remain so even when they make causal contact. This leads to a worldview that is highly counter-intuitive, but does not lead to conflicts with experience. Moreover, locality and compatibility with relativity theory are positive points of radical perspectivalism.

Keywords Perspectivalism · locality · covariance · relativity

1 Introduction: Unitary Quantum Mechanics and Perspectivalism

Traditional presentations of quantum mechanics assign a special role to the concept of measurement. According to this traditional view, the temporal development of an isolated system, or of an isolated collection of interacting systems, is described by unitary evolution in Hilbert space, while the effect of a *measurement* on such a physical system, or collection of systems, is taken to correspond to a non-unitary collapse of the quantum state.

Although this dualism of unitary and non-unitary evolution is part of standard textbook accounts, it is well known to raise fundamental problems. To start with, it is unclear what the defining difference is between measurements to be described by collapses, and interactions between object systems and measuring devices represented by unitary evolution. As perhaps first emphasized by Everett [1], even the consistency of the combination of unitary and non-unitary evolution may be questioned.

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Everett considered "nested experiments": an observer A performs a measurement on a system S while a second observer B, who is isolated from both A and S separately, is able to make measurements on the total system $A + S$. If the quantum state of S is a superposition of the eigenstates of the observable measured by A, the collapse postulate says that A's measurement outcome will be one of the eigenvalues of the measured observable, while as a result of the measurement intervention the state of S collapses to the corresponding eigenstate. By contrast, Everett argued, observer B will still consider $A + S$ as a system on which no measurement has been made, and whose evolution can be represented as a unitary process. From B's perspective $A + S$ will therefore end up in an entangled state, and not in a state corresponding to a definite outcome obtained by A (or a statistical mixture of such states). Consequently, Everett concluded, the collapse plus unitary evolution scheme of standard quantum mechanics leads to state assignments that are inconsistent with each other.[1]

A natural response is to remove the internal tensions in the formalism by stripping measurements of their special status and representing all interactions, regardless of whether they can be considered to correspond to measurements or not, in the same fashion. This leads to a theoretical scheme in which there is only unitary evolution: Unitary Quantum Mechanics (UQM). Clearly, in UQM, interactions between systems, including measurement interactions, generally lead to entangled states.

Experimental research of the last decades has lent empirical support to the hypothesis that UQM is the fundamental core of quantum mechanics: experiments have verified the existence of superpositions even when quasi-macroscopic objects are involved. Such experiments need to be sophisticated, since the omnipresence of decoherence in everyday circumstances masks the presence of superpositions by suppressing interference effects.

If UQM is accepted as a fundamental framework, the entangled states that arise as a result of measurement interactions must be reconciled with the experiential fact that only one specific measurement result materializes. We may take our cue from Everett again: Everett introduced the notion of *relative states*. The idea is that subsystems of a composite system that as a whole is represented by an entangled state do not possess pure states of their own, so cannot be in an eigenstate of the observable that is measured. However, these subsystems do have pure states *relative to states of the remainder of the total system*. For example, if the entangled state of $A + S$ is given by $c_1|o_1\rangle|A_1\rangle + c_2|o_2\rangle|A_2\rangle$, $|o_1\rangle$ is the relative state of S with respect to the state $|A_1\rangle$ of A, and $|o_2\rangle$ is the relative state of S with respect to the state $|A_2\rangle$ of A. Generally, if the entangled state of a composite system $X + Y$ is given by $\Sigma_i c_1|x_i\rangle|y_i\rangle$, X has the relative state $|x_i\rangle$ with respect to Y's state $|y_i\rangle$.

[1] By contrast, Wigner [2] later used the very same example to defend the view that collapses objectively take place as soon as the first *conscious* observer performs a measurement. Wigner's view has the weakness of invoking consciousness as a *deus ex machina*. It is possible in principle to test Wigner's idea, by checking whether the sealed lab containing A and S can be in a superposition. Induction on experimental results of recent years suggests that superpositions are maintained even at the macroscopic level, but it is true that no relevant experiments have been done with conscious observers.

Now, consider an ideal von Neumann measurement and a post-measurement relative state of system S with respect to a "pointer state" of the measuring device (a state that represents a record registered by the measuring device). Everett introduced the interpretative rule that in this case S possesses the property corresponding to the relative state (represented by the eigenvalue of the observable of which the relative state is an eigenvector) *in a relational way*. In the above example, S is characterized by the eigenvalue o_1 of the observable $\mathcal{O}$ with respect to A having registered outcome 1, and the value o_2 with respect to A having registered outcome 2.

Generally, *all* possible outcomes, correlated with their associated records, are represented in the post-measurement state. Everett emphasized that all these "branches" in post-measurement superpositions are equally "real" in the sense that their presence can be demonstrated, in principle, via experiments.

Consider, for example, the Wigner's friend scenario, in which an observer A in a sealed laboratory (let us call her Alice) performs a measurement as described above while B is located outside of the laboratory. B (Wigner) may decide to measure an observable of which $c_1|o_1\rangle|A_1\rangle + c_2|o_2\rangle|A_2\rangle$ is an eigenstate. In this way the external observer could verify, without disturbing the laboratory system, that there has been no collapse—that there still are two "branches" after the measurement by Alice.[2]

The theoretical scheme at which we arrive in this way is unitary and perspectival, with a perspectivalism that is grounded in physical facts [4]. Indeed, from the point of view of A, who has seen and remembers a specific outcome, system S must be described by the corresponding relative-state property—not as a matter of A's compelling subjective belief, but as the only correct description of the physical situation, which will be borne out by the results of experiments that A can perform on S. However, if B makes measurements on the lab containing A and S, he has to use the entangled state of $A + S$ in order to obtain empirically correct predictions. So from B's perspective, the lab has the physical properties associated with this entangled state.[3]

2 Radical Perspectivalism

Let us return to Alice who, inside her sealed laboratory, performs a spin measurement in the z-direction on an electron in the spin state $\frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle$. During the measurement interaction, the up and down spin states $|\uparrow\rangle$ and $|\downarrow\rangle$ become

[2] Everett's remark about the physical reality of all branches is usually interpreted as support for the idea of many worlds. This is contentious, however. There is historical evidence that Everett thought of an interpretation in which there is only one world, but with different perspectives defined within it. See [3], and references therein.

[3] We suppose here, and will suppose in the following, that a one-world interpretation can be given of the total state (comprising A, S and B) according to which *one* of the pointer states, or memory states of recording devices, is physically actual. The probability of being actualized is assumed to be given by the usual Born rule. The discussion of this assumption, which is common to one-world non-collapse interpretations, including the Bohm theory, Rovelli's relational interpretation, modal interpretations, and others, would go beyond the scope of the present article; see, e.g., [5–9].

correlated to Alice's memory, the notebook in which she writes down her result, and other objects inside the laboratory. Let the two macroscopically different total configurations of the lab and what is inside of it be represented by the orthogonal states $|L_\uparrow\rangle$ and $|L_\downarrow\rangle$, respectively. Then, assuming unitary evolution, the state of the lab after the measurement will be given by:

$$|\Phi\rangle = \frac{1}{\sqrt{2}}\big(|L_\uparrow\rangle + |L_\downarrow\rangle\big). \tag{1}$$

This state of the laboratory and its contents is a coherent superposition even though decoherence will have taken place inside the lab, so that for Alice it has become impossible to detect any interference between the branches. As we have already discussed, however, outside the lab an external observer (Wigner) is able to verify that the state assignment of Eq. (1) is correct, for example by measuring an observable of which this state is an eigenstate. The observable with the eigenstates $|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}\big(|L_\uparrow\rangle \pm |L_\downarrow\rangle\big)$ will do the job. Measurements of this observable will have the result associated with eigenvector $|\Phi^+\rangle$ with probability 1. By contrast, if the laboratory had been in a statistical mixture of $|L_\uparrow\rangle$ and $|L_\downarrow\rangle$, with equal statistical weights, this probability would have been only $\frac{1}{2}$.

But Wigner may also decide to make a completely different measurement on the lab, namely one whose result will show him what Alice has found in her measurement. He may simply open the door of the laboratory and ask Alice about her outcome. Or he might make a measurement on some object inside the lab that has become correlated with Alice's outcome. All measurements of this kind can be represented by an observable with the eigenstates $|L_\uparrow\rangle$ and $|L_\downarrow\rangle$. If we associate ± 1 with the two possible outcomes, the relevant observable takes the form

$$\mathcal{O} = |L_\uparrow\rangle\langle L_\uparrow| - |L_\downarrow\rangle\langle L_\downarrow|. \tag{2}$$

Now, standard discussions of the Wigner's Friend thought experiment assume that when Wigner measures $\mathcal{O}$, his outcome will simply reflect what Alice had already found when she did her experiment, before Wigner even considered making any measurements himself. This assumption may seem a self-evident piece of common sense. Nevertheless, the standard quantum formalism does not lend support to it [10, 11]. Indeed, standard quantum mechanics stipulates that when operator $\mathcal{O}$ is measured on a system in state (1), *two* outcomes are possible, $+1$ and -1, with equal probabilities $\frac{1}{2}$. Accordingly, it may happen that Alice has found "spin up", while Wigner finds that Alice, and all the things correlated with her, show evidence of "spin down". In other words, it may happen that from Wigner's perspective Alice has found "spin down", whereas from Alice's own perspective she has found "spin up". We are thus lead to the possibility of a *Radical Perspectivalism* (RP), according to which the relativization of facts with respect to observers persists even if these observers make causal contact and interrogate each other about their results.

This possibility of radical perspectivalism can only be removed by adding a new axiom to unitary quantum mechanics, according to which the probabilities of possible outcomes of measurements are not only determined by the quantum state

of the measured system, but also depend on determinate physical properties present in the system. In the case at hand, the suggestion is to add the rule that Wigner, upon entering the lab, must find the same results Alice had found earlier. It seems attractive to introduce such an axiom: indeed, one may even question the tenability of a radical dependence on perspectives and wonder about its compatibility with our experience of a shared world. We will discuss such arguments in Sect. 5.

First, however, we want to point out that, despite its weird appearance, radical perspectivalism possesses a number of conceptual virtues. As we will argue, RP makes it possible to give a *local* account of situations that are standardly considered as typically exemplifying quantum non-locality. Moreover, RP is able to circumvent a number of no-go theorems concerning the compatibility of unitary quantum mechanics and relativity theory. It is able to provide a consistent relativistically covariant interpretative scheme.

3 Radical Perspectivalism and Locality

To see how and in what sense radical perspectivalism is able to tell a local story in situations where Bell inequalities can be violated, consider the familiar EPR-Bohm setup in which Alice and Bob share a pair of electrons in the singlet spin state $\frac{1}{\sqrt{2}}(|\uparrow\rangle_A|\downarrow\rangle_B - |\downarrow\rangle_A|\uparrow\rangle_B)$, where the subscripts A and B refer to Alice and Bob, respectively. Both Alice and Bob possess a device that is able to measure spin in the z-direction; Alice's device is M_A and Bob's device M_B, respectively.

Suppose that Alice measures the spin of her electron and finds "up" as a result. The interaction between Alice's measuring device and Alice's electron is a local physical process, which creates a definite-valued spin in the z-direction, relative to Alice and her measuring device (the original singlet state was rotationally invariant). From the perspective of Alice (and her device, including records of the outcome) her electron has therefore acquired the definite spin value "up". Moreover, because of the strict anti-correlation that was present in the original singlet state, Alice now regards the spin of Bob's particle to be "down". In other words, a local event at Alice's position, comprising the spin measurement and its result, has locally affected A's perspectival description of the world. This change has the concrete consequence that if Alice travels to the other electron and measures its spin, she will certainly find "down". As we will further elaborate in a moment, this change in the properties of the electron at Bob's position, as defined from Alice's perspective, does not entail according to RP that something changes in that electron from Bob's perspective. So, it is not legitimate to conclude that something "in the electron itself" has changed due to Alice's distant measurement.

It is nevertheless true that the change in perspectival properties at a distance occurring here is peculiar from a classical viewpoint. The underlying new feature of quantum mechanics is that quantum mechanics allows the existence of definite correlations between quantities that themselves are not definite-valued. In our case, state (3) specifies a definite correlation between spins at the two wings of the experiment, while the spins themselves are not definite—there is no definite spin

direction associated with the electrons. This is different in classical physics, where physical quantities always are definite. To bridge the gap between classical intuition and quantum behavior to some extent, one might imagine that in the EPR-Bohm situation all anti-correlated spin instantiations exist jointly in potential form, and that Alice's local measurement intervention selects one of them. Another strategy for highlighting the local aspects of what is happening, is to view Alice's relative state not as representative of the *properties* of Bob's distant electron, but rather as a predictive tool for what Alice will find when she travels to the other electron and makes a spin measurement. In this case the anti-correlated spin value is only created when Alice interacts, locally, with the distant electron. This is the type of locality that Qbists claim for their approach (cf. [11]).

In order to clearly qualify as a non-local physical influence, something more is needed than the perspectival change we just described. Suppose that Bob measures the spin of his electron, in the z-direction, immediately after Alice has completed her z-spin measurement and has found "up". If Bob now must find "down" as his outcome, with probability 1, Alice's measurement has had a undeniable non-local influence. Indeed, before Alice's measurement intervention, Bob had two equiprobable possible outcomes, and immediately afterwards he has only one, so that he is restricted in his options. This non-locality is actually part and parcel of standard interpretations of quantum mechanics. However, it is not part of the scheme we are proposing here. According to radical perspectivalism *nothing changes* at Bob's position, from Bob's perspective, when Alice makes her measurement. Bob retains all the options he had.

To see why this is the case, consider the state of the two electron spins after Alice's measurement (as predicted by UQM):

$$\frac{1}{\sqrt{2}}(|M_A\uparrow\rangle|\uparrow\rangle_A|\downarrow\rangle_B - |M_A\downarrow\rangle|\downarrow\rangle_A|\uparrow\rangle_B). \tag{3}$$

Bob has to rely on this state in order to make correct predictions, in exactly the same way as Wigner had to rely on state (1) after his friend had performed a measurement in her laboratory. According to RP, the result of Alice's measurement (from her perspective) plays no role for Bob's predictions. So, Bob may find "spin up" or "spin down", both with probability $\frac{1}{2}$; Alice's result has no influence. The state of Bob's electron as defined by its density matrix has obviously also remained untouched by Alice's activities (this is the background of the no-signaling theorem).

So, Alice may find "up", from her perspective, while Bob from his perspective also obtains the result "up" when he measures his own electron. This shows why the perspectivalism that we discuss is radical: the two perspectives are completely unrelated. One may well wonder whether this independence of perspectives isn't in conflict with empirical evidence. Indeed, when we compare measurement records from the two wings of Bell-type experiments, we find correlations as predicted by standard quantum mechanics—these results are certainly not uncorrelated. But RP has no problem reproducing these well-verified predictions of quantum mechanics.

In order to investigate the correlations between the results on the two wings, we may introduce an observer who compares the outcomes of Alice and Bob. This new observer will obtain results that can be predicted on the basis of the state

$$\frac{1}{\sqrt{2}}(|M_A\uparrow\rangle|\uparrow\rangle_A|\downarrow\rangle_B|M_B\downarrow\rangle - |M_A\downarrow\rangle|\downarrow\rangle_A|\uparrow\rangle_B|M_B\uparrow\rangle). \tag{4}$$

The state (4) makes it obvious that our new observer, *from his perspective*, will find a perfect anti-correlation between A's and B's results. This is so, despite the lack of correlation between Alice's and Bob's results judged from their own perspectives.

We may also wonder what will happen when A and B meet, talk to each other and compare notes. This is similar to the question of what will happen when Wigner enters the laboratory of his friend and makes contact with her. Quite generally, it can be shown that no contradictions will arise in such situations and that all quantum predictions will be confirmed within each single perspective. In Sect. 5 we will say more about the overall picture that arises in this way.

4 Radical Perspectivalism and Relativistic Covariance

Interpretations of quantum mechanics have a difficult relationship with the principle of relativity. Collapse versions of quantum mechanics (in which measurements cause the wave function to collapse) face the problem that the collapse is standardly assumed to take place instantaneously over a simultaneity hyperplane. In addition to the problem that this is a superluminal process, there is also the problem that there are as many simultaneity hyperplanes as there are inertial frames. So, which hyperplane should be chosen? Non-collapse schemes turn out to experience similar difficulties. In Bohmian mechanics, for example, one cannot assume that all inertial systems are equivalent. For this reason Bohmians accept the existence of a preferred frame of reference. Several no-go theorems [12–14] show that other interpretations of unitary quantum mechanics are vulnerable to essentially the same problem.

The core difficulty, on which the no-go theorems capitalize, is that one would like measurement results and other properties of systems, and their probabilities, to be unambiguously given, independent of the simulaneity hyperplane on which they are described. But the no-go theorems demonstrate that such descriptions cannot always consistently be given in the case of measurements of the EPR-type.

Gao [15] has illustrated this tension between unitary quantum mechanics and relativity by means of a simple thought experiment. In order to understand its idea one should realize that it is a feature of unitary quantum mechanics that local measurements can in principle be undone, also locally, through a reversal of the local measurement interaction. Needless to say, such a reversal, including the erasure of records and the undoing of all decoherence processes that have taken place, is practically impossible—but we are discussing matters of principle here, relating to the structure of unitary quantum mechanics.

So, suppose that Alice performs a spin measurement on her part of a pair of electrons in the singlet state, and that her local measurement is reversed and undone

a bit later. Bob, at spacelike separation from Alice, measures the spin of his electron, in the same direction as Alice. Because of the spacelike separation, it depends on the frame of reference whether Bob's measurement takes place before or after Alice's experiment has been undone. Standard reasoning now implies that Bob's result has either to be opposite to what Alice has found or is independent of it, depending on the frame of reference and its simultaneity hyperplane from which the prediction is made. Indeed, if Bob's measurement is later than Alice's but earlier than the reversal of Alice's measurement, Bob's outcome must be opposite to Alice's outcome. But if Bob's measurement takes place after Alice's measurement and outcome have been undone, his result may be either "up" or "down", with equal probabilities.

Applying this argument to a long series of repeated measurements leads to a logical contradiction. If all measurements in the series are considered in which Alice finds "up", Bob must in all these cases find "down" if we use a definition of simultaneity according to which Bob's measurement takes place before Alice's measurement will be undone. In contrast to this, Bob should find roughly equally many "ups" and "downs" according to a definition of simultaneity that makes his measurement later than the reversal of Alice's measurement. Note that this contradiction is not empirical: it is not a contradiction that could be noted by Bob as some kind of absurdity in his measurement results. Indeed, Bob has no means to select measurement events, on his side, that correspond to Alice's outcome "up". Still, the contradiction signals an inconsistency. So, there must be something wrong with the premises of the argument.

The introduction of Radical Perspectivalism makes this contradiction disappear. As explained in Sect. 3, RP predicts that it makes no difference at all for Bob's possible outcomes and their probabilities if we conditionalize on Alice making or not making a measurement, or on the undoing or not of the measurement. In all these cases Bob, from his perspective, has two possible outcomes, with equal (conditional) probabilities. This is exactly what we should expect on the basis of relativity theory.

It is true that from Alice's perspective, after her measurement, Bob's electron has acquired a definite spin, opposite to the one of her own electron, and that she expects to verify that value if and when she travels to Bob. In our just-discussed thought experiment this perspectival view of Alice disappears together with her measurement outcome when the measurement interaction is reversed.

Scenarios are obviously possible (and much more normal) in which Alice's measurement is not undone, and in which she actually travels to Bob and measures a spin value there that is opposite to her earlier measurement result, see Sect. 3. According to RP, this measured spin of Bob's electron, from Alice's perspective, does not need to coincide with what Bob measures, from his own perspective. The situation is analogous to what happens, according to RP, when Wigner enters the previously sealed laboratory and asks his friend what the result of her spin measurement was. In that case, the answer received by Wigner need not correspond to what Wigner's friend actually found. So, according to RP there may be conflicting descriptions of a physical system, defined with respect to different perspectives.

The RP account therefore strongly differs from the standard analysis, which says that the outcome of a measurement is an objective fact, accessible—in principle—to all observers.

5 Sharing Facts in Radical Perspectivalism

An immediate concern is that RP leads to contradictions; or at least to a picture in which observers are trapped within their own perspectives so that communication with others is precluded. If the latter were the case, no scientific communities could be formed, making science impossible. RP would thus erode the basis under its own existence [16].

But the idea that RP leads to contradictions or to solipsism is too quick. It is of course true that when Wigner goes into his friend's laboratory, asks about her result and gets another answer than her own previous measurement result, there is no longer one coherent world characterized by unique measurement outcomes. But this strange feature does not show up in empirical results within a perspective and does not lead to contradictions. The conflicting facts are components of different perspectives. Wigner could investigate further, and start collecting evidence on Alice's response. Everything he will find in the laboratory will confirm Alice's statements with a probability of 1, as can easily be seen from the conditional probabilities calculable in state (1). What is more, repeated investigations may reveal physical regularities to Wigner, and could motivate him to formulate general scientific hypotheses. This all happens within his own perspective. But the regularities, laws and theories that he may discover transcend this perspective [11, 17]: these general scientific results are valid from *all* perspectives. Therefore, even though Wigner cannot access the private perspectives of his colleagues, he can have fruitful exchanges with colleagues within his own perspective. The theories that are born from this cooperation, within a perspective, will be valid within all perspectives.

Although RP does not conflict with any empirical data and does not cause obstacles for communication and science, it must be admitted that it is a counter-intuitive position. Objective states of affairs that are basically accessible to everyone are part and parcel of the common-sense worldview. One would therefore perhaps expect that the perspectives of RP would merge, perhaps approximately, in limiting cases, particularly in everyday circumstances. Since decoherence plays a major role in suppressing the detectability of quantum effects in the macroscopic world, it may seems plausible to expect that decoherence mechanisms will also alleviate the conceptual threats of perspectivalism. Indeed, it has been suggested, in the context of discussions of relational/perspectivalist interpretations of quantum mechanics, that a common decoherence domain produces shared facts [17, 18]. But it is not immediately clear here what this "sharing" should mean, and what the exact role of decoherence is. Consider the following story.

Alice, in her sealed lab, has performed her spin measurement. She finds herself in a decoherent environment within her lab. This environment couples, with (practically) orthogonal states, to Alice's two outcome states. Outside of the lab is

Wigner1 and his clone Wigner2. Wigner1 opens the door of the lab, so that part of the decoherent atmosphere escapes and becomes even more inaccessible (and decohering) than it already was. First Wigner1, and then Wigner2 goes in and takes a look at Alice's log. The successive stages of this process can be represented as follows, in obvious notation:

$$\begin{aligned}
&(c_1|A_\uparrow\rangle|E_\uparrow\rangle + c_2|A_\downarrow\rangle|E_\downarrow\rangle)\,|W_1(0)\rangle\,|W_2(0)\rangle && \rightarrow \\
&(c_1|A_\uparrow\rangle|E_\uparrow\rangle|W_1(\Uparrow)\rangle + c_2|A_\downarrow\rangle|E_\downarrow\rangle|W_1(\Downarrow)\rangle)\,|W_2(0)\rangle && \rightarrow \\
&(c_1|A_\uparrow\rangle|E_\uparrow\rangle|W_1(\Uparrow)\rangle W_2(\Uparrow)\rangle + c_2|A_\downarrow\rangle|E_\downarrow\rangle|W_1(\Downarrow)\rangle W_2(\Downarrow)\rangle).
\end{aligned} \tag{5}$$

According to radical perspectivalism, the meaning of these equations is as follows. Alice will have found, in her perspective, either "up" or "down", with respective probabilities of $|c_1|^2$ and $|c_2|^2$ (first stage). Then Wigner1 enters and investigates the scene. According to RP, he may also find "up" or "down", regardless of Alice's perspectival outcome. So Alice"s result may be different; nevertheless, from Wigner1's perspective, Alice will have found the same result as he did. Finally, Wigner2 comes in. His result will again be either "up" or "down". From Wigner2's perspective there is complete agreement between his own information and that obtained by Alice and Wigner1. Complete mutual agreement between Alice, Wigner1 and Wigner2 also obtains within the other two perspectives, although these other perspective-internal agreements may be about different measurement results.

It is therefore true that within each perspective shared facts are created, so that normal community life is possible [17]. In this, decoherence processes play the relatively minor role of stabilizers, guaranteeing that the different perspectives remain separate and do not interfere.[4] It would be wrong to suppose that decoherence will align the facts defined in different perspectives, creating a stockpile of "absolute" facts.

As is clear from the above story, according to RP it may well happen that different perspectives assign different perspectival properties to a given system. Wigner may even be justified in assigning properties to Alice that differ from the properties she experiences herself. But such differences cannot be detected by empirical means within any perspective. Everything measured by Wigner will confirm the property attributions defining his own perspective (and will of course confirm the supra-perspectival rules of quantum mechanics); Alice's measurements will do the same with respect to her perspective. So, the hypothesis that Wigner and Alice do not share the same facts is not empirically testable. This might be seen as a motivation to regard this hypothesis as "meaningless" [17] and reject it on this ground.

The qualification "meaningless" does not seem appropriate, though: the *meaning* of RP is clear enough and may be given a formal underpinning by supplementing each equation line of (5) with the actual recorded measurement results. But the complaint may nevertheless be taken seriously in the form of the adage "do not

[4] That is, the perspectives remain separate as long as everything evolves irreversibly. A further caveat is that an *external* super-observer who could monitor everything, including the complete environment E, would be able to verify that the total system is in a superposition of perspectives.

accept more metaphysics than what is needed to account for what can be observed." However, this maxim is controversial. It is not unusual at all in physics to have different theoretical schemes that are equally capable of doing justice to observable phenomena—the various interpretations of quantum mechanics are a good example. Debates surrounding such schemes focus on *theoretical virtues*, like compatibility with fundamental ideas of relativity theory, the avoidance of superluminal interactions, and so on. And as we have seen, RP offers advantages on these points.

6 Summary and Conclusion

There are good physical reasons to think that unitary quantum mechanics constitutes the fundamental core of quantum mechanics. As first discussed by Everett, this unitary formalism naturally evokes the notion of "perspectives". Everett provided a formal tool for defining such perspectives, namely his relative states. In this paper we have followed Everett's example and have used relative states to establish perspectival descriptions with respect to a given physical system.[5]

We have focused on a radical form of perspectivalism, in which it may be true that Alice, from Wigner's viewpoint, has noted "down" in her log, while Alice herself, from her perspective, knows and sees that she has noted "up". This lack of shared facts between perspectives is highly counter-intuitive. However, it does not lead to contradictions. The relative state formalism shows that each single perspective is completely consistent and satisfies the principles of quantum mechanics. Each perspective is internally "normal": it conforms to the usual rules of physics and does not show traces of what is going on within other perspectives.

Nevertheless, the introduction of this radical version of perspectivalism smacks of a redundant proliferation of metaphysics. However, surprisingly, Radical Perspectivalism is able to tell a local story of what happens in experiments of the EPR-type. Moreover, it offers a Lorentz covariant description of such experiments. Therefore, RP can avoid the introduction of a privileged inertial system of reference. It is true that other frameworks, e.g. the Bohm theory, achieve what might be called "empirical Lorentz covariance". The privileged frames in empirically Lorentz covariant schemes are not detectable by empirical means, so that one may select any inertial frame to serve as the special one in which the fundamental laws in their standard form are valid. However, one may well argue that this strategy of introducing a watered-down notion of compatibility with relativity, according to which it must be accepted that not all inertial frames are equivalent, conflicts with the spirit of relativity theory. RP respects this spirit of relativity. However, its radical perspectivalism conflicts with the spirit of common sense. There is a trade-off here

[5] This way of defining perspectival descriptions differs, although perhaps only on a minor point, from the relational approach introduced by Rovelli [5]. Rovelli insists that relative (perspectival) facts are only created in interactions, and do not exist between non-interacting systems. So, for example, the properties of Alice's lab are not factual with respect to Wigner before he interacts with it. This is different in the relative state approach.

between different theoretical virtues, and it is not trivial to decide whether RP fares better or worse than other empirically adequate interpretations.[6]

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[6] A question we have not addressed here, but which seems very relevant, is whether the different perspectives of RP can be considered different *worlds* and whether this would make RP close to, and perhaps even indistinguishable from, the many-worlds interpretation.

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